

Error Pattern Analysis: Negation Scope Detection with CRF

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January 30th, 2025

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Introduction

- Systematic error analysis based on CRF classifier results for negation scope detection given the cue.
- Error sample selection:
 - Sampled 50 random tokens with false positive or false negative predictions.
 - Identified 20 error categories based on:
 - Annotation of negation cues and their scope Guidelines
 - Universal Dependency Relations

Error Pattern Analysis

MC_and_SC (Frequency: 6)

- **Description:** Negation cue in the main clause, root in scope, subordinate clause also in scope.
- **Guideline:** If main clause semantically affects subordinate clause, the latter should be in scope.

Errors:

Subordinate clause missing in predictions

1. **Prediction:** [There 's] **no** [link between the two save for Henderson 's own servant], who serves the family 's meals.
2. **Gold:** [There 's] **no** [link between the two save for Henderson 's own servant, who serves the family 's meals].

Subordinate clause depends on another subordinate clause (negation in latter)

1. **Prediction:** I was sure Henderson, as he calls himself, felt that he was suspected, and that [he would lie low and make] **no** [move so long as he thought he] was in any danger.
2. **Gold:** I was sure Henderson, as he calls himself, felt that he was suspected, and that [he would] lie low and [make] **no** [move so long as he thought he was in any danger].

Error Pattern Analysis

Sub_CL_only (Frequency: 5)

- **Description:** Negation affects only the subordinate clause, not the clause it depends on (main or another subordinate).
- **Example:**

Prediction: Telegraphic inquiries had been already made [which showed that Marx knew] **nothing** [of his customer save that he was a good payer].

Gold: Telegraphic inquiries had been already made which showed that [Marx knew] **nothing** [of his customer save that he was a good payer].

Error Pattern Analysis

Pred_exp_OOS (Frequency: 5)

- **Description:** Sentences where a predicative expression following a copulative verb are incorrectly marked as out of scope by the CRF classifier.

- **Example:**

Prediction: [Your cipher was] **not** difficult, madam.

Gold: [Your cipher was] **not** [difficult], madam.

Error Pattern Analysis

Aff_skipped (Frequency: 4)

- **Description:**

- The negation cue is an affix attached to an adjective or adverb.
- Scope is incorrectly marked as empty by the CRF classifier.
- Two common cases:
 - Adjective refers to the subject → only the subject is within scope.
 - Adjective refers to something else → entire clause is within scope.
- For adverbial negation cues, the scope spans the entire clause.

- **Example:**

Prediction: “Remarkable, but by no means **im**possible,” said Holmes, smiling.

Gold: “Remarkable, but by no means **im**[possible],” said Holmes, smiling.

Counterfactual Analysis

MC_and_SC Analysis

- **Identification:**

- Used spaCy dependency labels: **ROOT**, **ccomp**, **relcl**, **advcl**.
- Conditions:
 - Negation cue in main clause.
 - Root in scope.
 - Sentence contains a subordinate clause in scope.
 - Predictions lack root or subordinate clause.

- **Example:**

Prediction: [We have] **no** [excuse for an intrusion upon his privacy until we have some reason] to think that there is a guilty reason for it.

Gold: [We have] **no** [excuse for an intrusion upon his privacy until we have some reason to think that there is a guilty reason for it].

- **Results:**

- **Correct sentences:** 36
- **Incorrect sentences:** 19
- Root missing: 6
- Subordinate clause missing: 13
- Error rate: ~ **34%**

- **Impact:**

- Semantic relations hard to capture by CRF model.
- Complex feature interactions (e.g., token distance, dependency path).

Counterfactual Analysis

Pred_exp_OOS Analysis

- **Identification:**

- Used **acomp** to detect adjectives following a copulative verb.
- Conditions:
 - Adjective is part of the scope.
 - Adjective appears in the predictions column.
 - Accounted for adjectives containing negation cues.

- **Example:**

Prediction: “[Your case is] **not** complete, then?” I asked.

Gold: “[Your case is] **not** [complete], then?” I asked.

- **Results:**

- Sentences with acomp in scope and predictions: **44**
- Incorrect acomp predictions: **9**
- Error rate: **~16%**

- **Impact:**

- Moderate performance, but scope prediction for predicative expressions needs improvement.

Conclusion

CRF → Great overall performance

Semantic complexity in subordinate clauses and predicative expressions often led to errors.

Majority of cases handled correctly, but error patterns indicate limitations in semantic generalization.

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